DATA ANALYSIS REPORT ON STUDENT'S SOCIAL MEDIA ADDICTION

Submitted By

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INTRODUCTION

Objective

The goal of this project is to understand how social media addiction impacts students' academic, emotional, and social well-being through the use of data-driven techniques. By performing Exploratory Data Analysis (EDA) on real-world survey data, we aim to uncover meaningful patterns, relationships, and trends between students' social media usage, addiction scores, academic performance, mental health, and interpersonal relationships.

- Quantify the prevalence of social media addiction among students.
- Examine how excessive social media use correlates with lower academic performance.
- Assess whether addiction levels are linked to deteriorating mental health scores.
- Evaluate how social media engagement contributes to relationship conflicts and changes in social behaviour.

Dataset Overview

The dataset contains 705 rows and 13 columns, it includes demographic details such as age, gender, academic level, and country, along with variables that measure daily social media usage, addiction levels, mental health scores, and relationship status. it also records students' perceptions of whether social media affects their academic performance and if it causes conflicts in their relationships.

- student_id: unique identifier assigned to each student respondent.
- most_used_platform: identifies the social media platform most frequently used by the student, such as Instagram, Facebook, Snapchat, or Twitter.
- Age: represents the age of the student.
- Gender: indicates whether the student is male, female, or other.
- Academic_Level: specifies the student's level of study.
- Country: the country of the respondent.
- Avg_Daily_Usage_Hours: the average number of hours a student spends daily on social media.
- sleep_hours_per_night: represents the average hours of sleep a student gets each night, which may indicate the effect of social media usage on sleep patterns.
- Addicted_Score: measures the extent of social media addiction.
- Mental_Health_Score: indicates the mental health condition of the student.
- Affects_Academic_Performance: shows whether social media has impacted academic results.
- Relationship_Status: describes the student's relationship situation.
- Conflicts_Over_Social_Media: indicates if social media has caused relationship conflicts.

Data preprocessing

Data cleaning

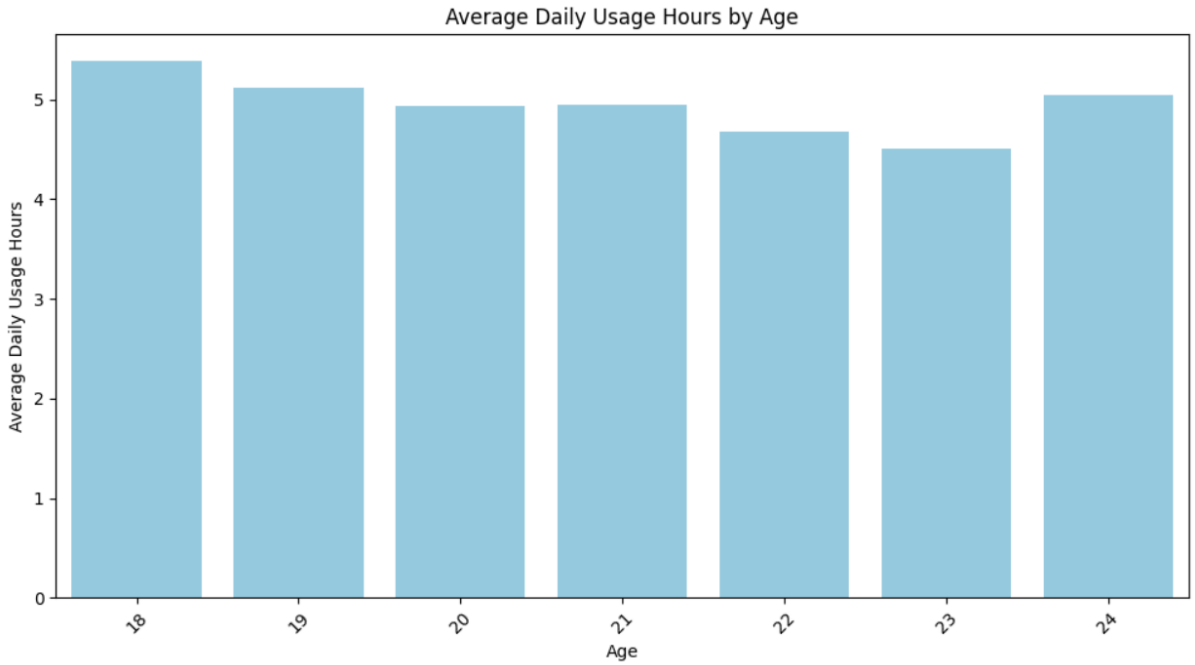
- Handling missing values: any rows with missing or incomplete data were reviewed
- Removal of unnecessary Columns: The dataset contained an “Student_ID” column, which was removed as it did not contribute to the analysis.
- Verified and adjusted data types for each column to ensure accuracy during analysis.
- Filled the missing values in numeric columns with median and replaced categorical columns with mode.

Exploratory data Analysis (EDA)

AGE

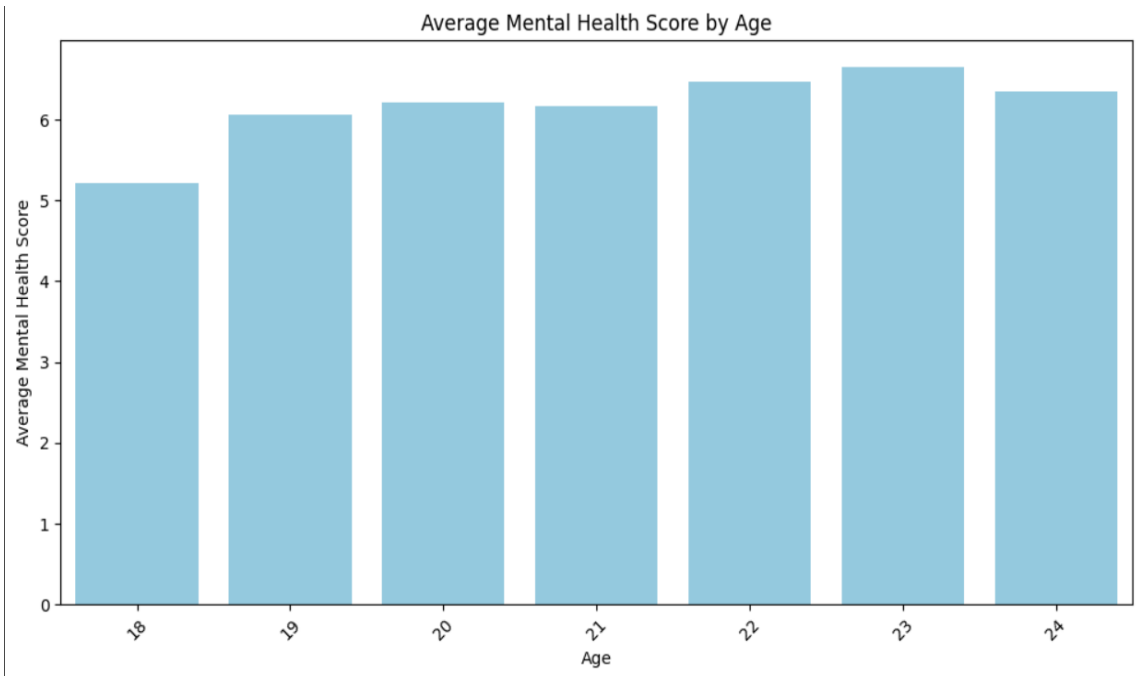
Group by age and calculate average daily usage

AGE	
18	5.385714
19	5.120245
20	4.930303
21	4.950641
22	4.676190
23	4.508824
24	5.04615



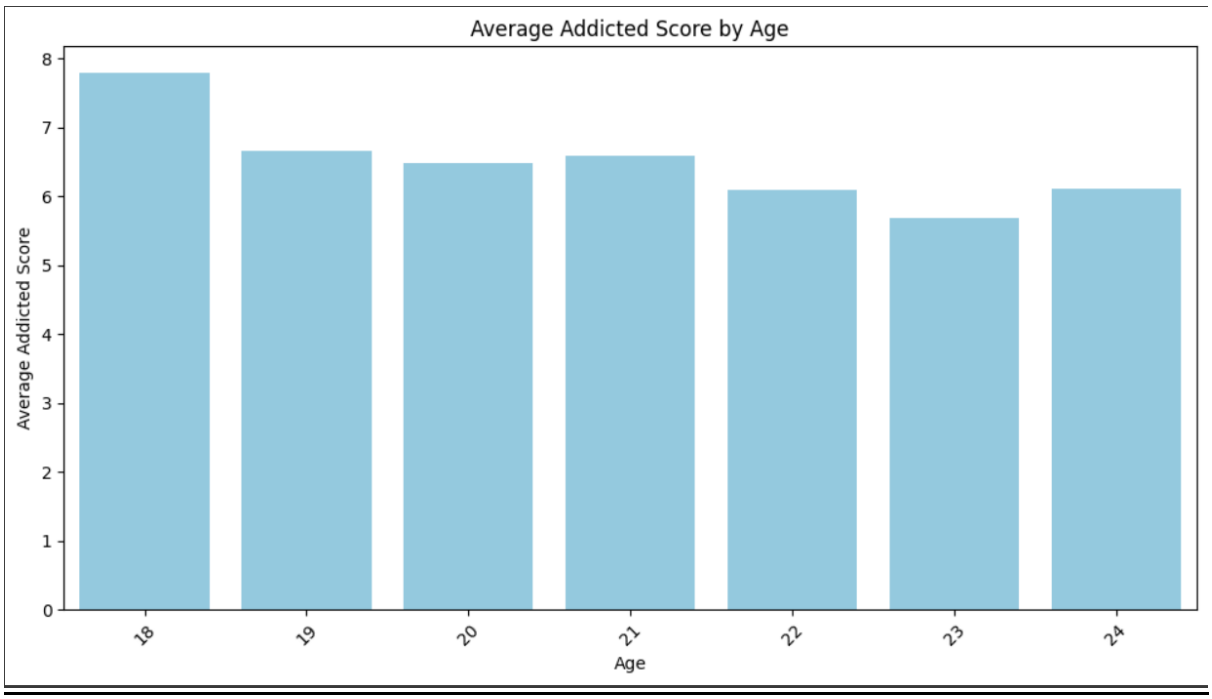
- Students aged 18 spend the most time on social media, averaging around 5.4 hours per day.
- As age increases, the average usage gradually declines, reaching its lowest at around age 23 (approximately 4.7 hours per day).
- However, there is a slight increase again at age 24, suggesting that social media engagement remains significant even among slightly older students.

Average mental health score per age



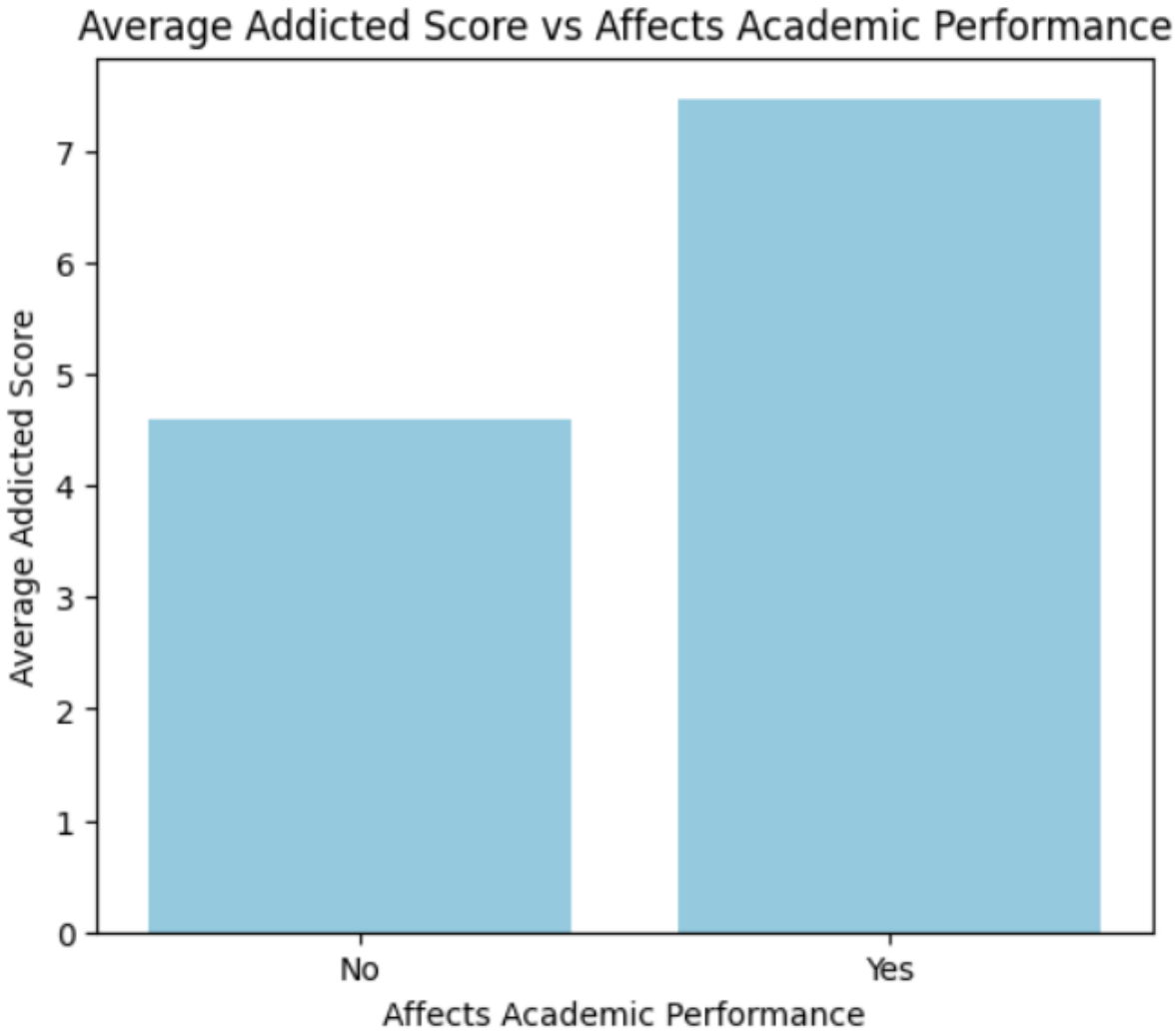
- Age 18 has the lowest average score (around 5.2).
- Scores generally increase with age, peaking at Age 23 (around 6.6).

Average addicted score per age



- Age 18 has the highest average addicted score, at almost 8.0.
- The scores generally trend downward, with the Age 23 group having the lowest score, at approximately 5.6.
- In short, younger participants (18-year-olds) report the highest levels of addiction.

Average addicted score and affects on academic performance



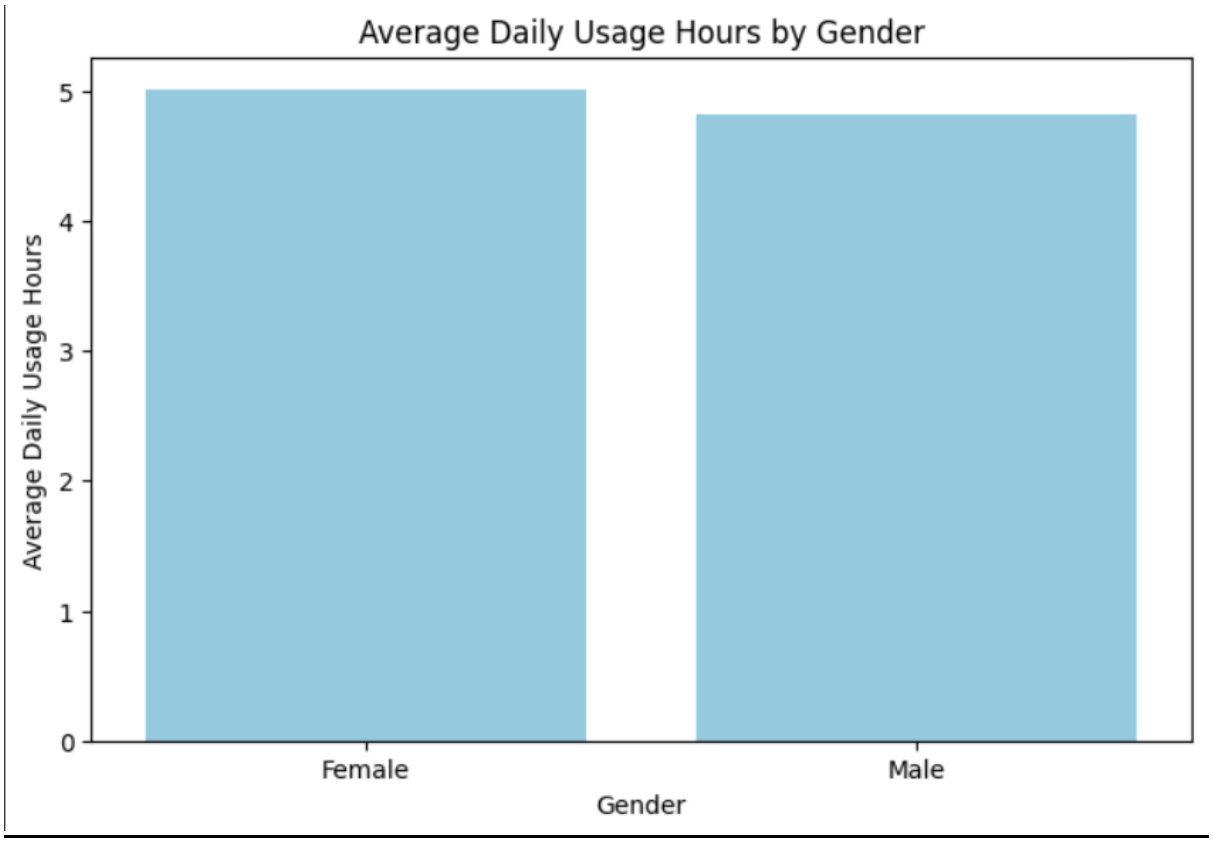
- Individuals who report that their usage affects their academic performance ("Yes") have a much higher average addicted score, at approximately 7.4.
- Individuals who report no academic impact ("No") have a significantly lower average addicted score, at approximately 4.6.
- In short, a higher addicted score is strongly associated with self-reported negative effects on academic performance.

Conclusion: Younger students (Age 18) report the lowest average mental health scores and affects on academic performance. Students aged 18 are at the highest risk profile, exhibiting the highest usage, the highest addicted score, low academic performance and the lowest mental health score. Academic performance generally deteriorates as addiction to social media

increases. The combined data strongly suggests a negative correlation between intensive social media engagement and well-being.

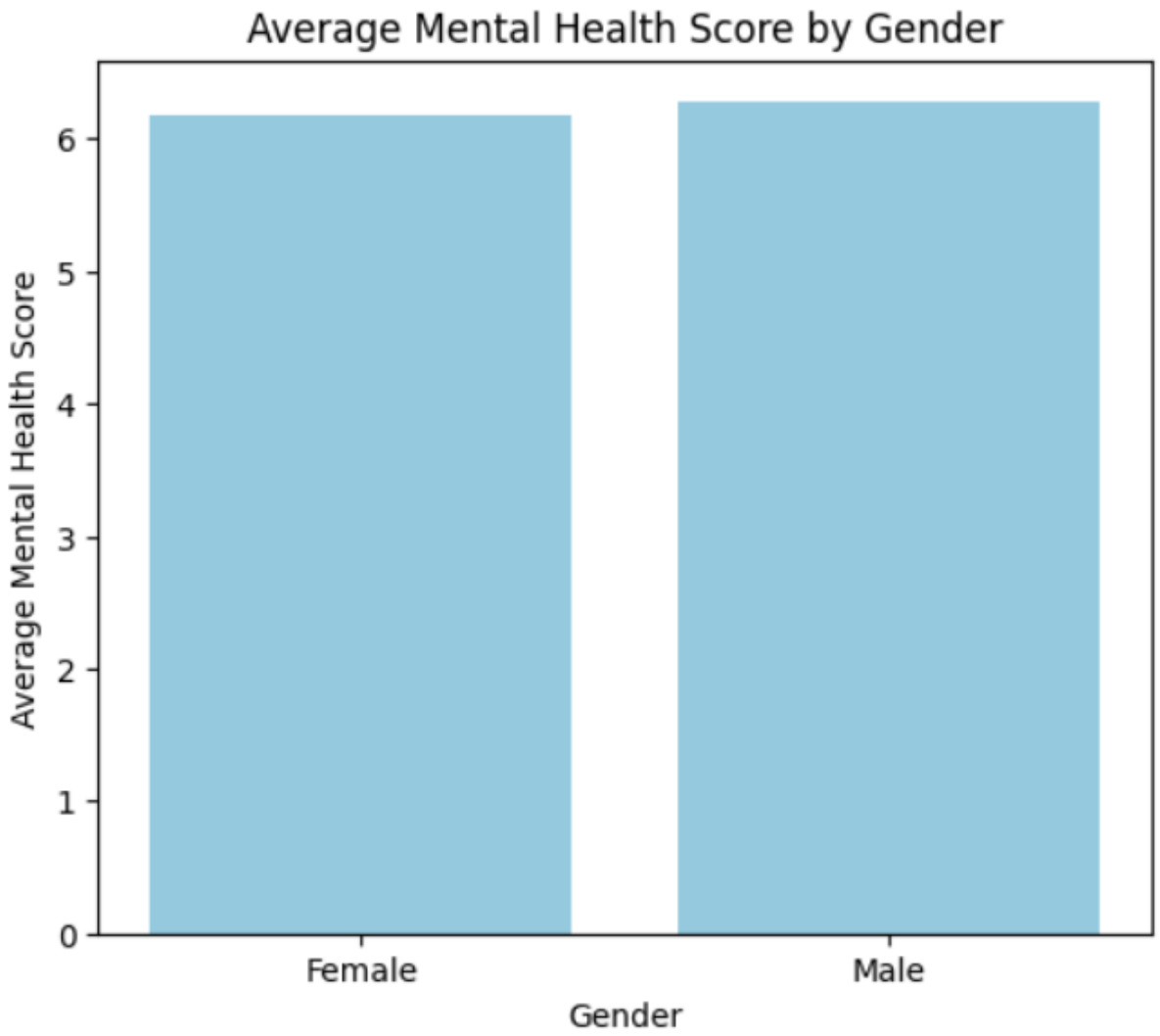
GENDER

Average daily usage per gender



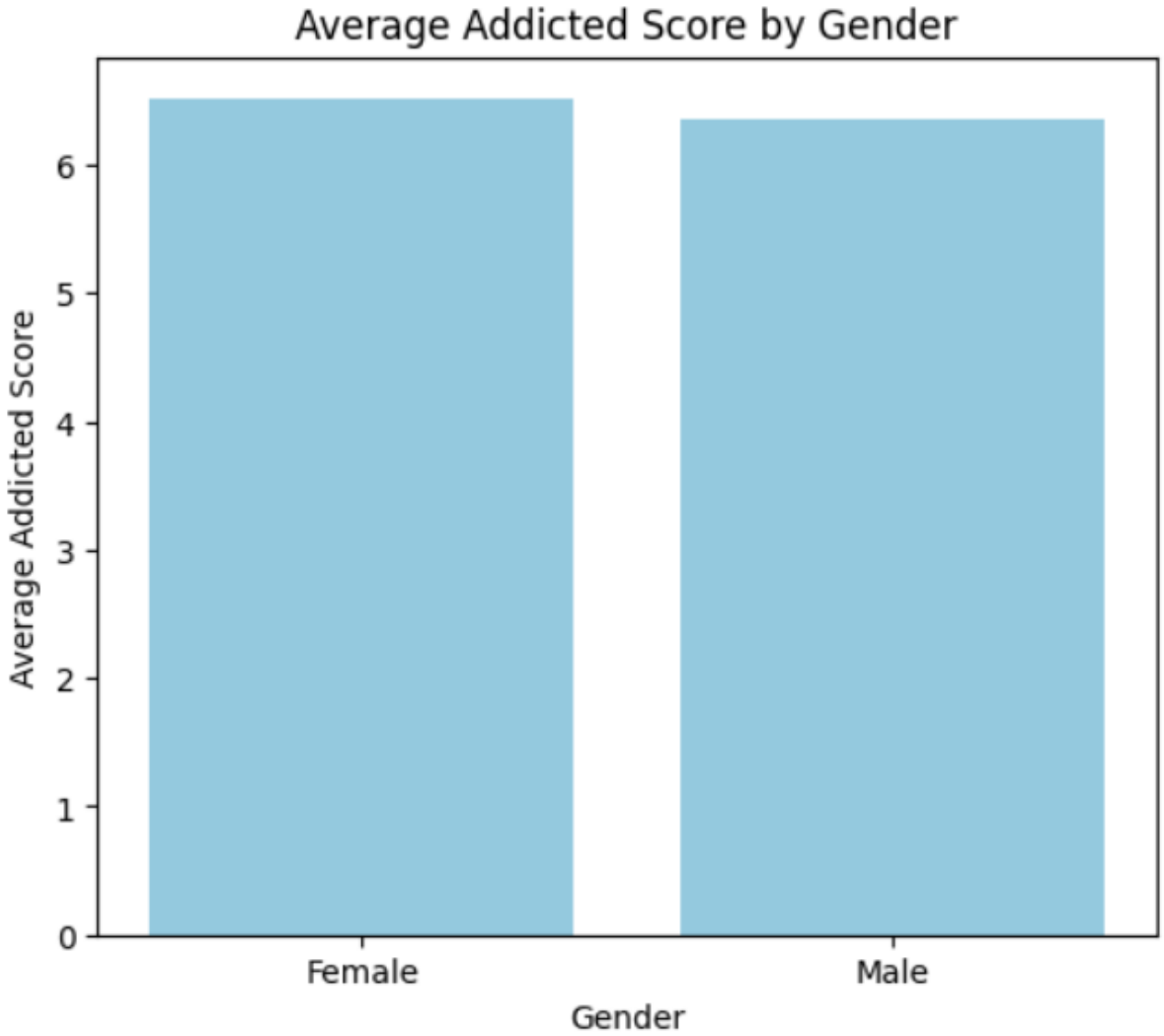
- Females have a slightly higher average usage, at approximately 5.0 hours per day.
- Males average slightly less, at about 4.8 hours per day.
- In short, both genders in this sample spend almost the same amount of time on social media daily.

Average mental health score per gender



- Females have an average score of just over 6.1.
- Males have a score that is negligibly higher, also just over 6.1.
- In short, there is virtually no difference in the average mental health score between the genders in this dataset.

Average addicted score per gender



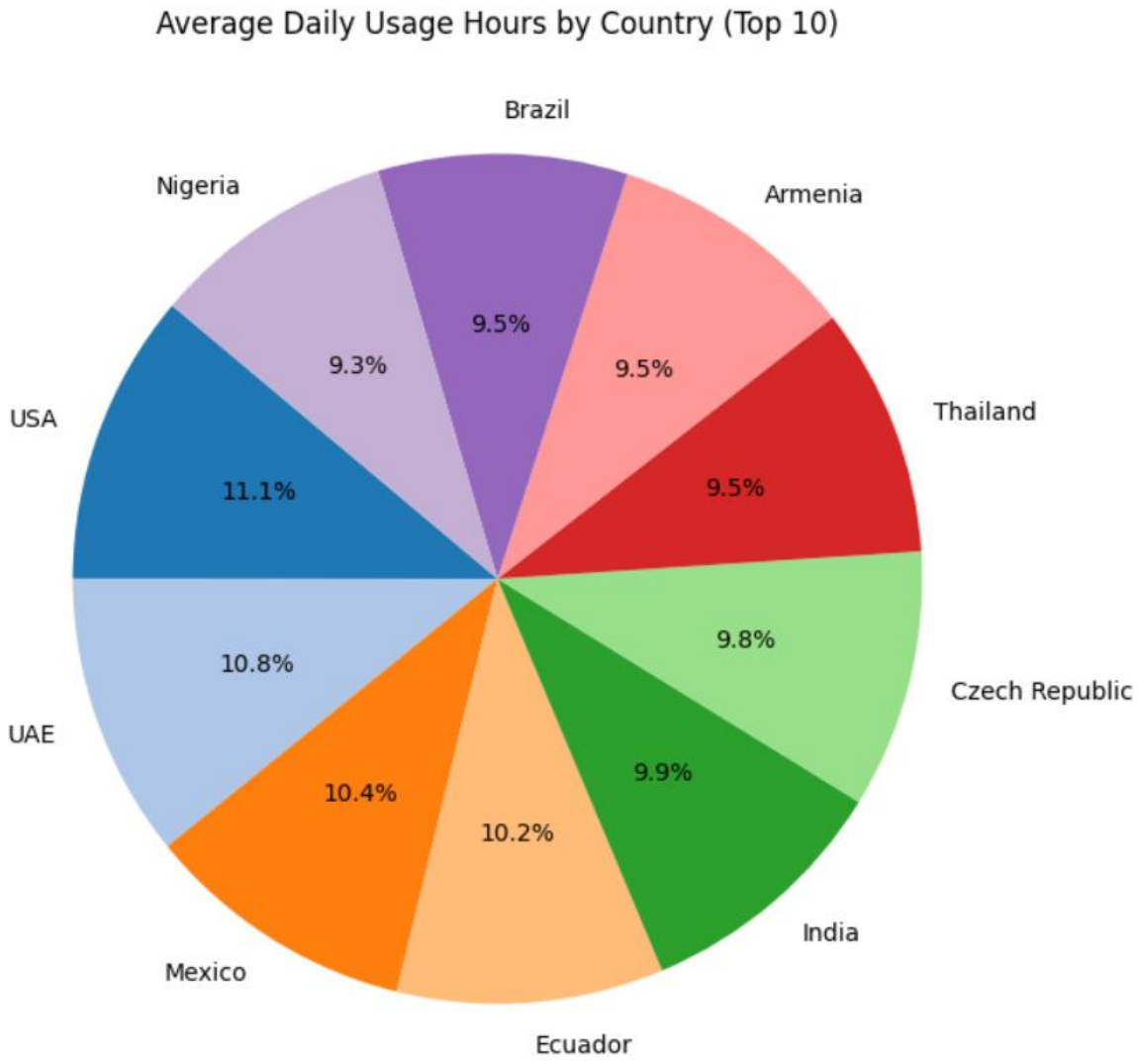
- Females have a slightly higher average addicted score, at approximately 6.4.
- Males have an average score that is negligibly lower, around 6.3.
- In short, gender is not a significant differentiator for the average addicted score in this sample.

CONCLUSION: Females have slightly higher averages for both daily usage (5.0 hours vs. 4.8 hours) and addicted score (6.4 vs. 6.3). Conversely, males have a negligibly higher average mental health score (both scores are just over 6.1). In summary, gender is not a significant factor in differentiating average daily

usage hours, addicted scores, or mental health scores in this dataset, as the averages are very close for both groups.

COUNTRY

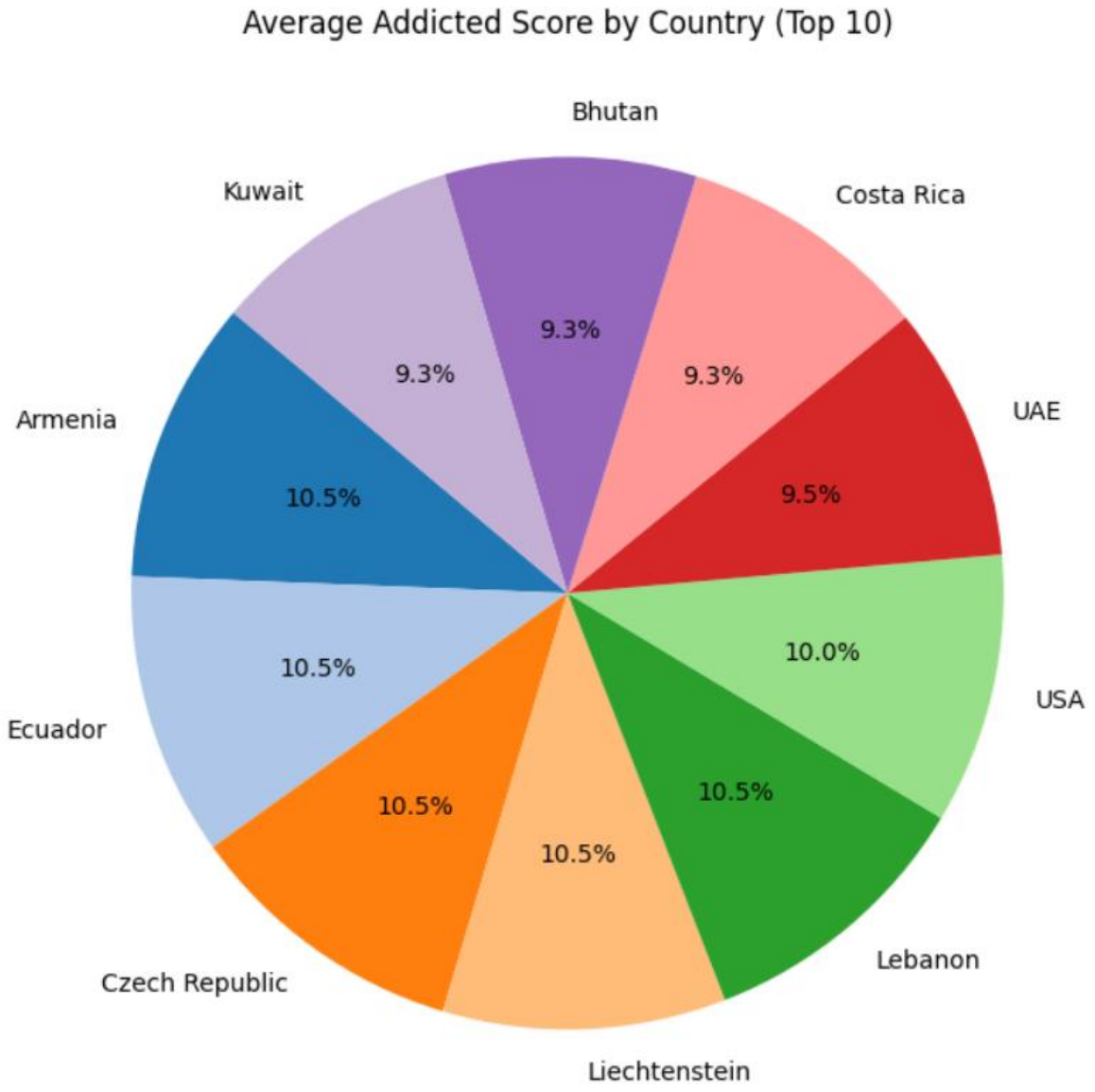
Average daily usage per country



- The **USA** contributes the highest proportion, accounting for 11.1% of the total usage shown.
- The **UAE** and **Mexico** are the next highest, contributing 10.8% and 10.4%, respectively.

- The remaining seven countries—Ecuador, India, Czech Republic, Thailand, Armenia, Brazil, and Nigeria—all contribute very similar proportions, ranging from 9.3% to 9.9%.
- In short, the average usage hours are highly distributed across the top 10 countries, with the USA being the slightly largest contributor.

<u>Average addicted score per country</u>

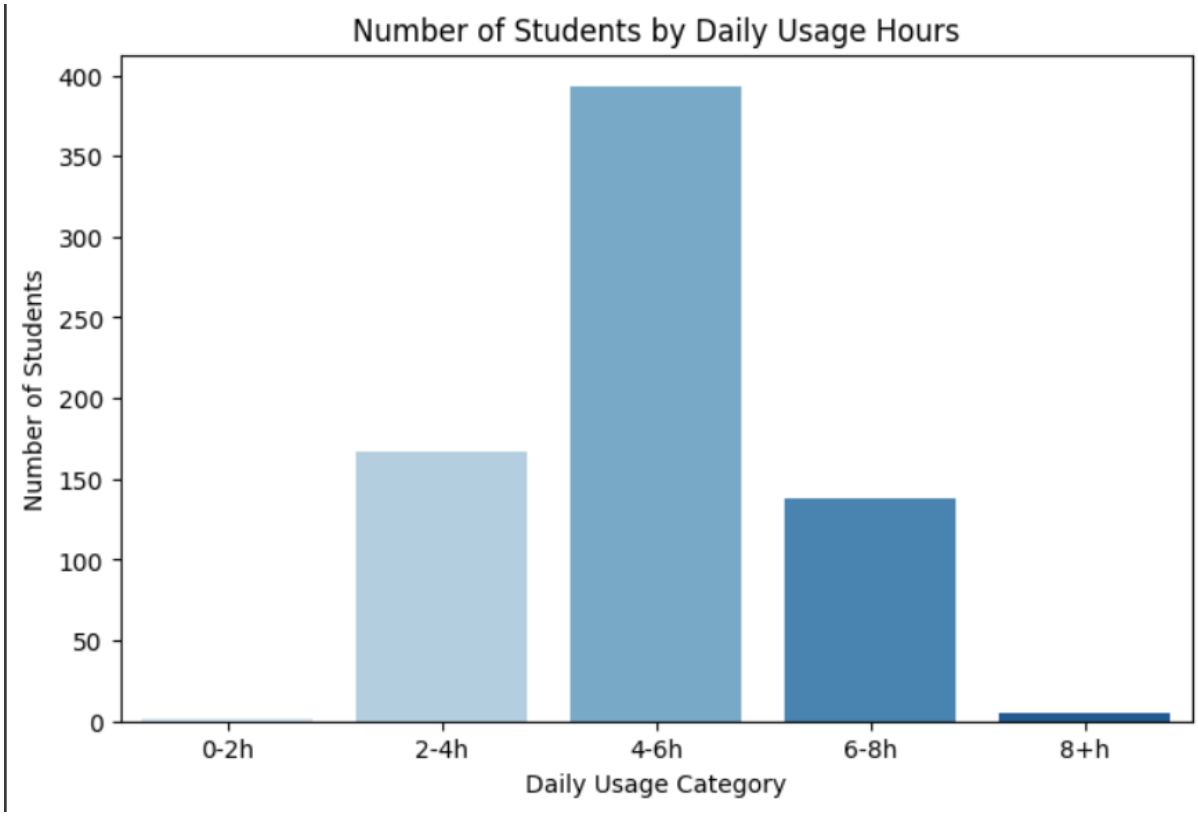


- Four countries—Armenia, Ecuador, Czech Republic, and Liechtenstein—share the highest proportion of the total addicted score, each contributing 10.5%.
- The USA and Lebanon follow closely, contributing 10.0% and 10.5%, respectively. (Note: The visual proportion for Lebanon appears slightly

smaller than the 10.5% share listed for the others, but the percentage is identical).

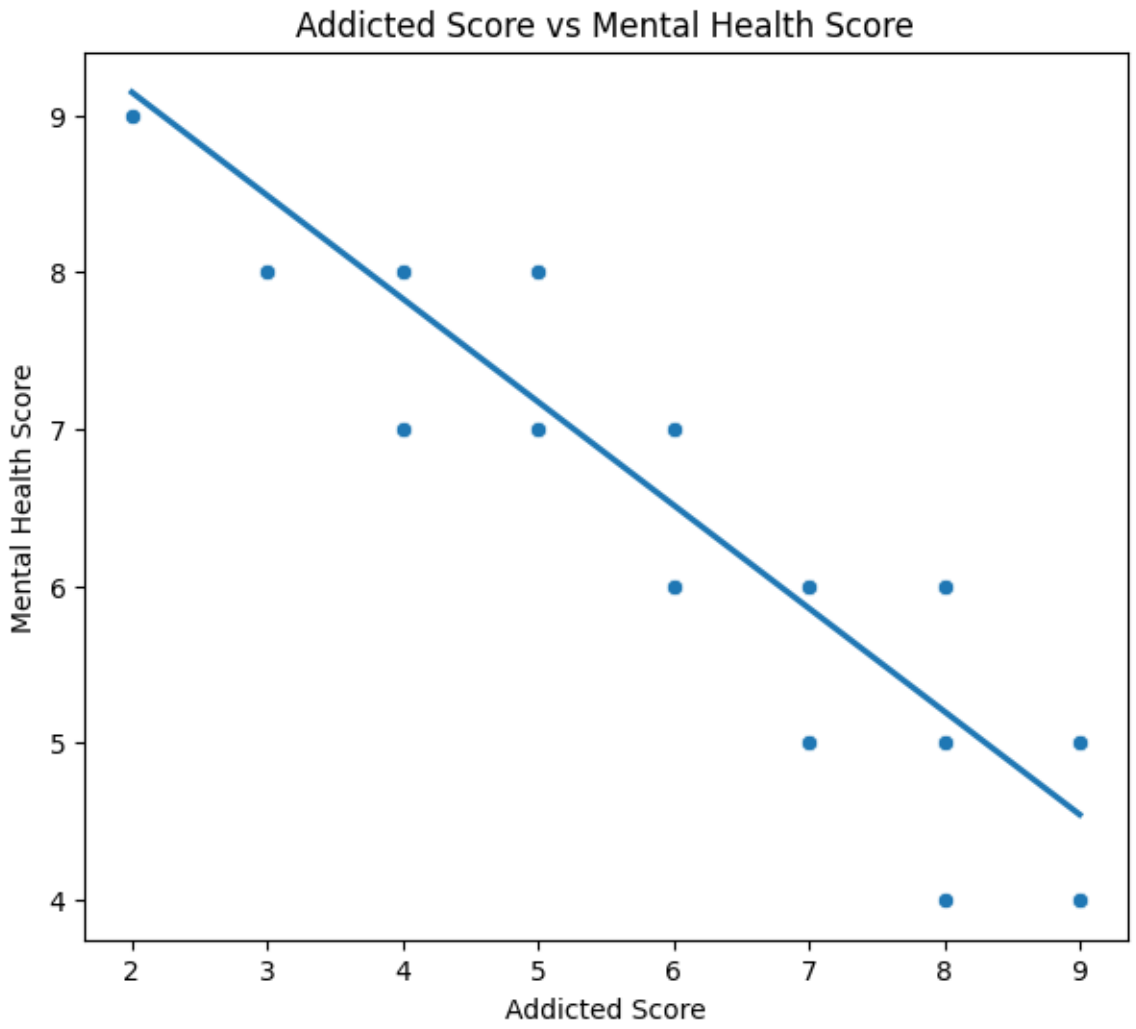
- The remaining countries (UAE, Costa Rica, Kuwait, and Bhutan) all contribute a lower, but very similar, share (9.3% to 9.5%).
- In short, no single country strongly dominates the average addicted score; the total is highly dispersed among these top 10 nations.

Create categories for usage hours



The vast majority of students fall into the 4–6 hours daily usage category, with nearly 400 students in this group. The 6–8 hours category is the second largest, while very few students use social media for less than 2 hours or more than 8 hours daily.

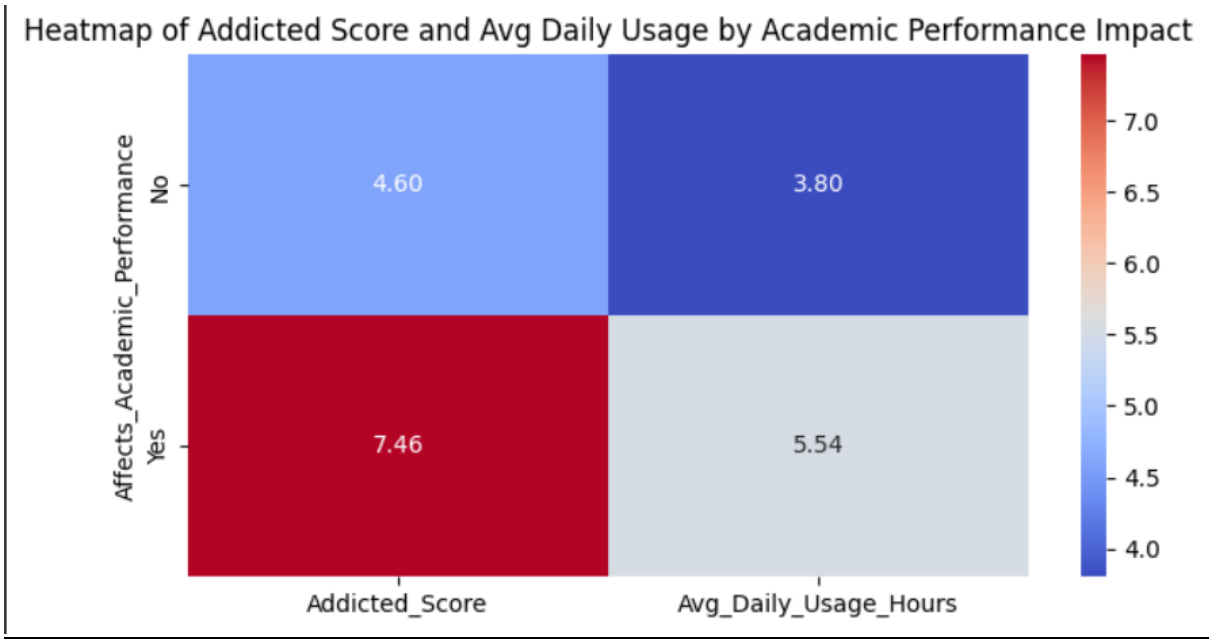
Addicted score vs mental health score



- The graph shows a clear negative correlation between the two variables, which means that as one score increases, the other tends to decrease. Specifically, as the Addicted Score increases along the horizontal axis, the Mental Health Score generally decreases along the vertical axis.
- The data points are closely grouped around the descending straight line, indicating a strong relationship.

- Individuals with a low Addicted Score (e.g., 2 or 3) tend to have a high Mental Health Score (e.g., 8 to 9). Conversely, individuals with a high Addicted Score (e.g., 8 or 9) tend to have a low Mental Health Score (e.g., 4 to 6).

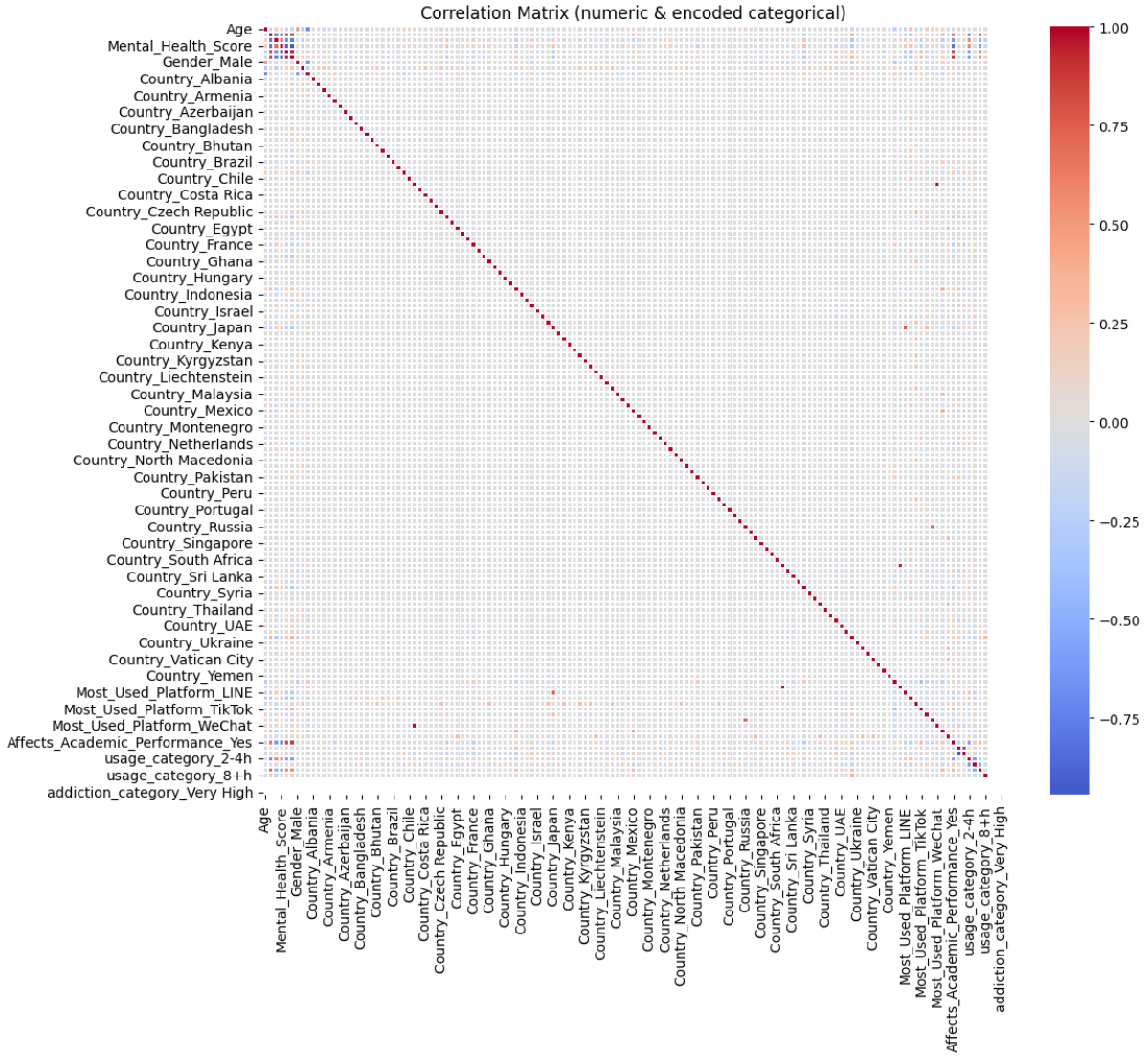
Heatmap of addicted score and average daily usage by academic performance impact



- Students who report that their usage affects their academic performance ("Yes") have a significantly higher Average Addicted Score (7.46) and higher Average Daily Usage Hours (5.54 hours).
- Students who report no academic impact ("No") have a much lower Average Addicted Score (4.60) and lower Average Daily Usage Hours (3.80 hours).

CONCLUSION: The difference in social media usage between genders is minimal, but the severity of addiction and the total time spent using social media are the key factors strongly correlated with negative academic outcomes. Students whose academics are affected report both much higher addiction scores and substantially longer daily usage times than their peers.

Correlation matrix



Weak Relationships: The graph contains light colours (white and very light red/blue), meaning that for pairs of factors (like Age and Country or Gender and Mental Health Score), the relationship is very weak or non-existent.

Colour Key: Dark Red means the two factors tend to increase together (strong positive link).Dark Blue means as one factor increases, the other tends to decrease (strong negative link).

Stronger Relationships: Stronger correlations (darker dots) are scattered, but appear most frequently among the last few variables listed, such as:

1. **Affects Academic Performance**
2. **Usage Category**
3. **Addiction Category**

CONCLUSION

Addicted Score and Academic Performance

- There is a strong connection showing a negative impact on academic performance.
- The data clearly indicates that individuals who report their addiction Affects Academic Performance have a significantly higher median Addicted Score (around 7.5) compared to those who report no impact (around 4.5).
- This suggests that a higher addiction level is correlated with students feeling their academic success is compromised.

Addicted Score and Mental Health

- There is a strong negative relationship between Addicted Score and Mental Health Score.
- As the Addicted Score increases, the Mental Health Score generally decreases.
- This suggests that higher levels of addiction are strongly associated with poorer mental health.

Average Usage Hours

- The overall correlation matrix indicates that most variables, including Usage Hours (implicitly), have a weak relationship with the other factors.
- However, external research suggests that while average total usage time correlates with addiction, the *type* of usage (e.g., communication/social apps vs. utility apps) and the *appraisal* of that use are often better predictors of addiction than just the total hours spent.

Synthesis of Key Relationships

In simple terms, the data points to a cascading negative effect:

1. A higher Addicted Score is strongly linked to worse Mental Health.
2. A higher Addicted Score is also strongly linked to the feeling that one's Academic Performance is negatively affected.
3. While a simple increase in Usage Hours is not the primary driver for all other issues, the quality of that usage, especially as it contributes to the Addicted Score, appears to be a critical factor in both mental health and academic outcomes.

Excessive social media usage quickly translates to a high addicted score, which is strongly associated with poor mental health. This high addiction is the key differentiator between students whose academics suffer and those whose do not.